# Pin and Replay, or Refuse: Exact Incremental Re-annotation Across Reference Database Releases

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## Abstract

Biologists annotate genomes by searching their proteins against reference databases such as Pfam, which publish a new release every few weeks to months. Each release forces a full re-search, although most of the database has not changed: 86.6% of Pfam's entries were identical between releases 38.1 and 38.2. Reusing the old results is hard, because the search programs are fixed binaries whose printed statistics depend on the size of the whole database. We present TARE (Tool Annotation Reuse, Exact), a wrapper that reruns an unmodified program only on the entries that can matter, with the program's own database-size setting pinned to the new release, so that the program itself prints the numbers a full run would print. TARE then tests the replay by adding back entries it left out, and refuses, returning an ordinary full run, if any line changes. When it ships, every output line except comment headers is byte-identical to a full rerun; the assumption that makes this possible is tested on every update, not proved.

On Pfam updates, TARE is 4.24× faster than a full rerun (median of nine genomes on a dedicated server, 95% CI 4.17–4.68). Across 715,704 output lines from HMMER and Infernal, not one differed from a full run. TARE refused DIAMOND, whose first filter reads bytes of the neighboring database entry, and BLAST+, whose duplicate output rows our parser cannot yet key.

## 1 Introduction

When a laboratory sequences a genome, it obtains thousands of protein sequences with no labels. To learn what each protein does, researchers compare it against *reference databases*: curated collections of known protein families, such as Pfam [18], UniProt [26], and InterPro [19]. A program such as HMMER [11] scores every protein against every family and reports the good matches, called *hits*. This step, called *annotation*, underlies much of modern biology, from comparative genomics to the sequence alignments used by structure predictors such as AlphaFold [13].

Reference databases keep improving. UniProt and InterPro publish a new version, called a *release*, roughly every eight weeks [19, 26], and Pfam published eight releases between May 2024 and June 2026 [21]. Each release adds and revises families, so annotations computed against an old release go out of date. The annotation programs keep no memory of earlier runs, so the only way to update is to compare every protein against the entire new database again. Most of that work recomputes answers that did not change: between Pfam 38.1 and 38.2, 86.6% of the 30,134 families were byte-for-byte identical (Figure 1). At scale this repeated cost shapes practice. NCBI re-annotates its RefSeq collection on a rolling schedule because redoing all of it at once "requires securing at once a significant amount of computing resources" [15], and JGI states that at over 65 billion genes "it is impossible to upgrade all genome and metagenome annotations to the latest version" [5].

**Why reuse is hard.** It might seem easy to search only the new and edited families and keep the old results for the rest. It is not, for three reasons. First, the results are *coupled*: the statistic HMMER prints for each hit, its E-value, is scaled by the total number of families in the database, much as a grade curved to class size changes for every student when new students enroll. Adding families therefore changes the printed numbers of hits on families that did not change. Rerunning HMMER on exactly the right families, without accounting for this, changes 9,083 of 9,094 output lines for one bacterial genome. Second, the programs are unmodified binaries maintained by others; we cannot change their code. Third, an approximately correct answer is not good enough: pipelines that track provenance must reproduce a full run exactly, because "pipeline differences can lead to annotation discrepancies" [5].

Existing tools fall short in two ways. General-purpose caches, such as build systems and workflow managers [7, 10, 17, 24, 28], remember whole commands keyed by the bytes they read. A new release changes the database file, so they rerun the search in full. Specialized incremental search tools such as iBLAST and iSeqSearch [9, 29] are written by hand for one program and adjust the printed numbers after the fact.

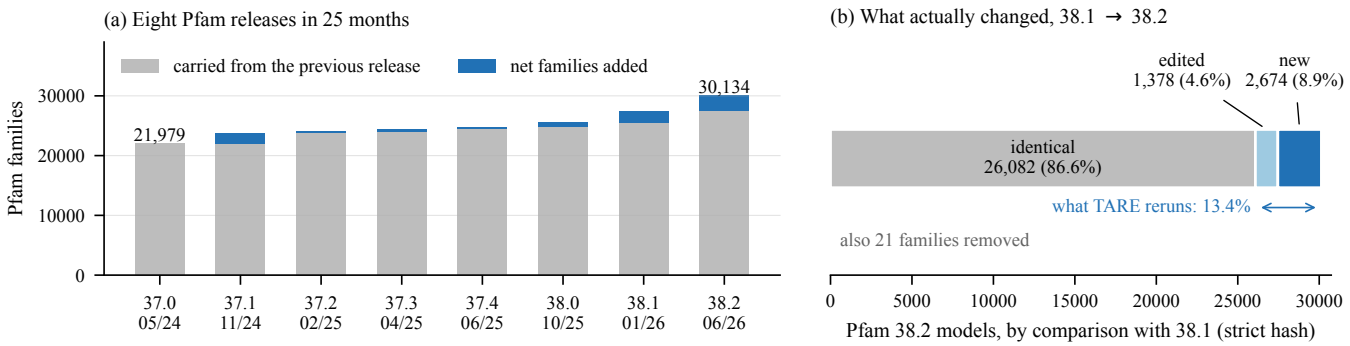



Figure 1: **Reference databases change often, but usually only in part.** (a) Pfam grew from 21,979 to 30,134 families over eight releases; blue segments show the net families each release added. (b) Comparing every model of Pfam 38.2 with 38.1 byte for byte: 86.6% are identical. Only 13.4% (new or edited) can produce new results, yet a conventional update recomputes 100%. Blue marks what TARE reruns. Panel (b) is the pair we time; Figure 6 measures the six neighboring releases the same way (median changed fraction 12.2%).

Their output is approximate, and they cannot tell when their adjustment is wrong: in our experiment, only 1.3% of the rows that iSeqSearch shared with a full run were byte-identical, and it reported 133 hits to sequences that the new release had deleted. *How can we reuse earlier annotations across a database release, without modifying the programs, and either return exactly what a full rerun would print or knowingly fall back to one?*

**The key idea in plain terms**

In the programs we studied, entries in the database influence each other's printed results through only a few global numbers, chiefly the database size, and the programs already accept those numbers as settings. If we tell the program the new release's size up front (*pin* it), it scores each family on its own merits. Then we can rerun it on a small subset of the database and it prints exactly what a full run would print, digit for digit, because the program, not our wrapper, computes every number. This is what *taring* a scale does: fix the baseline first, then weigh only what is new. When a program breaks this pattern, we try to detect it and fall back to a full run.

Pinning a global size is not new in itself. Parallel search tools split a database across machines and fix the database size so that every piece prints consistent E-values [8], and HMMER's `-Z` option exists partly for that purpose [11]. What is missing is a way to use it across releases, for a binary we do not control, without trusting that the binary has no other coupling. TARE supplies that: it finds which entries changed (Section 3), reruns each protein only against its few candidate families with the size pinned (Section 4.5), and before returning anything adds back entries it left out and refuses if any line changes (Section 4.6). Every refusal falls back to an ordinary full run, so users always receive the program's own output.

We evaluate TARE on four programs: two where it reproduces full runs exactly (HMMER and Infernal) and two it refuses: DIAMOND, whose results depend on database order, and BLAST+, whose output our parser cannot yet key. On Infernal, updates ran about twice as fast as a full rerun; on genomes made of a few long DNA sequences, TARE skips two of its sampled checks and relies on the remaining two (Section 6.3). Our contributions are:

1. **Exact replay across releases with check-or-refuse.** Release-diff discovery and candidate-only, batched, pinned replay that makes an unmodified program print byte-identical output, guarded by a probe-based gate and a check, either of which refuses whenever the replay could differ (Section 4).
2. **A root cause for a refusal.** Investigating a failure of our first check revealed why DIAMOND cannot be replayed exactly: its first filter reads bytes belonging to the neighboring database entry, so its results depend on database order (Section 6.4).
3. **A pre-registered evaluation.** Exactness over 715,704 output lines, speedups with their run-to-run variation, known-bad controls that must refuse, and comparisons with general-purpose caches and approximate incremental search, from experiments whose design and pass/fail rules were committed before they ran (Section 6).

## 2 Background

### 2.1 How Annotation Works

An annotation program takes two inputs: a set of *records*, such as the thousands of proteins of a bacterial genome, and a *reference* of *entries*, such as Pfam's 30,134 protein families (Table 1). Each Pfam family is stored as a statistical profile, a profile hidden Markov model, that summarizes what members of that family look like. HMMER's `hmmscan` compares every protein with every family, computes a score, and writes one line per reported hit, containing the names of the

Table 1: Terms used in this paper.

| Term | Meaning |
|---|---|
| Record | One query item, here one protein (or one DNA sequence for Infernal). |
| Reference | The database searched against, here Pfam or Rfam; a file of *entries*. |
| Entry | One family model in the reference (a profile HMM in Pfam). |
| Release | A new version of the reference. |
| Hit / row | A reported record–entry match; one line of the output table. |
| E-value | How many matches this good would be expected by chance in the whole database; scaled by the database size $Z$. |
| Stock run | Running the unmodified program on everything, the result we must reproduce. |
| Pin | Fixing a global setting (e.g., $Z$) to the new release's value. |
| Replay | Rerunning the unmodified program on a subset of the reference. |

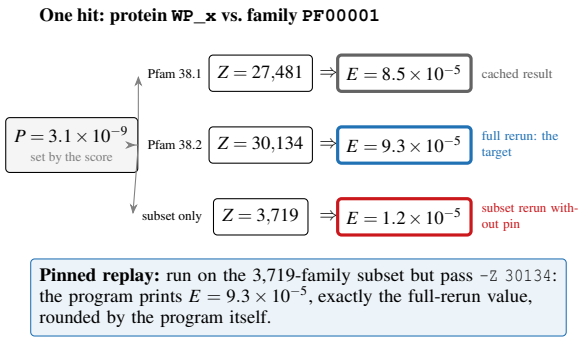



Figure 2: **Why old results cannot simply be reused.** An E-value is a probability $P$ (fixed by the score) times the database size $Z$. The same hit prints a different E-value in each release, and a different one again if the program is rerun on a subset. Pinning $Z$ makes the subset run print the full-run value. (Illustrative $P$; $Z$ values are Pfam 38.1, 38.2, and one proteome's candidate subset.)

protein and family, the score, and an E-value. The E-value estimates how many equally good matches would occur by chance in a database of this size, and HMMER computes it as a probability multiplied by $Z$, the number of families in the database [11]. With Pfam's curated per-family score thresholds (the `--cut_ga` option), whether a hit is reported depends only on its score against that family's threshold [18].

## 2.2 Why Naive Reuse Fails

Suppose a laboratory annotated a genome with Pfam 38.1 and wants Pfam 38.2 results. The obvious shortcut is to search only the 4,052 families that are new or edited and keep the cached lines for the other 26,082. Nearly every kept line is then wrong, because its E-value was multiplied by 38.1's $Z$ (27,481) instead of 38.2's (30,134) (Figure 2). Rerunning HMMER on exactly the right families does not help either, because the program then uses the subset's size as $Z$: on one genome, such a rerun differed from a full run on 9,083 of 9,094 lines, and the other 11 lines matched. Multiplying the cached E-values by $Z_{\text{new}}/Z_{\text{old}}$ gets closer but still fails, because printed numbers are rounded, and the rounding of a rescaled number differs from the program's rounding of its own internal value.

# 3 Problem Model

The effect of a release travels along a chain: a release changes some entries, and changed entries change some output lines. The goal is to produce every line of the new full run while running the program on as little as possible. We make this precise.

**Definition 1** (Release)**.** A reference $R$ is a list of entries in file order. Each entry has a name and a *strict hash* $h(e)$ of the content that affects scoring. A new release $R'$ splits into *stable* entries $S = \{e \in R' : h(e) \in h(R)\}$ and *changed* entries $\Delta = R' \setminus S$.

**Definition 2** (Exact re-annotation)**.** Let $f(Q,R)$ be the program's output on records $Q$ and reference $R$, and $\text{body}(\cdot)$ the output with comment lines removed. Given $f(Q,R)$ and $R'$, an exact re-annotation returns $O$ with $\text{body}(O) = \text{body}(f(Q,R'))$ byte for byte, or refuses and returns a full run $f(Q,R')$.

**Definition 3** (Coupling)**.** A program is *coupled* across entries if the line for record $q$ and entry $e$ can depend on entries other than $e$. We distinguish four forms: a *normalizer*, a global number computed from the whole reference, such as $Z$; a *side file*, an extra input file that lists entry names; an *entry group*, a set of entries whose hits compete, such as Rfam's clans; and *neighbor coupling*, where the program reads bytes of the entries next to $e$ in the file.

**Definition 4** (Entry locality under a pin)**.** A program is *entry-local under pin* $\pi$ if, for every record $q$ and every subset $U \subseteq R'$ that contains all entries reported for $q$ in the full run, running it on $(\{q\}, U)$ with its normalizers set to $\pi(R')$ prints exactly $q$'s lines from the full run.

Entry locality is what makes exact replay possible. It is a property of the program and its options, not something TARE proves; TARE tests it before and during every replay and refuses when a test fails. We assume the program is deterministic for fixed options and thread count, and that every input it reads is named on its command line. TARE treats the program as a black box at run time and never reads its source code. We, the authors, did read the source code of HMMER, Infernal, and DIAMOND, to design experiments and to explain their outcomes; none of that knowledge is built into TARE.

## 4 Design

TARE updates a cached annotation in six stages (Figure 3). The first three decide *whether* exact replay is possible and *how*; the last three perform it and check it. Any failed decision or check leads to a full stock run.

### 4.1 Stage 1: Entry Diff

TARE splits both releases into entries and computes a *strict hash* of each: SHA-256 over the entry's full text, including its name, accession, description, length, Pfam's curated thresholds (GA, TC, NC), and every model line, after dropping only lines tagged DATE (build date), BM and SM (the commands that built and calibrated the model), or COM (comments). A renamed or re-thresholded family therefore counts as changed. Entries of the new release whose hash appears in the old one are stable; all others are changed. A cache keyed on the whole database file would miss on every entry after a release; a cache keyed on entry hashes keeps every stable entry, 86.6% of Pfam for 38.1→38.2.

### 4.2 Stage 2: Contract Inference

TARE must learn how the program's output depends on the reference without reading the program's code. It runs the program on a sample of records three times: against the whole reference, and against each half of it. It then lines up the output lines of the three runs by record and entry name.

*Does the output split cleanly?* If every line of the whole run appears in exactly one half-run, the output *decomposes* over entries: each line comes from one entry. If the whole run reports a line that neither half reports, entries interact in ways TARE cannot replay, and it refuses. The opposite case, a half reporting a line the whole omits, happens when entries compete and suppress each other; TARE accepts it only when the user declares which entries compete (Section 4.3).

*How does each number depend on the reference?* For each numeric column, TARE tests a small fixed set of hypotheses: the column does not depend on the reference at all (*identity*), or it scales with the number of entries, their total length, or the number of hits per record. A hypothesis fits if rescaling every half-run value by the corresponding ratio reproduces the whole-run value within the printed precision. A column that fits no hypothesis causes refusal. Columns that are ranks or labels rather than measurements are recognized and left to the program to reprint.

The result, called a *contract*, records how the output splits and how each column depends on the reference. It is computed once per command and reused for every later release.

### 4.3 Stage 3: The Coupling Gate

The gate turns the contract into a decision. Users may add declarations, which are data, never program names.

*Size dependence → pin.* If a column scales with the reference size, TARE needs a program option that sets that size. Users declare it (`--pin '-Z {entry_count}'`), and every experiment in this paper used a declared pin. TARE can also search for one, by reading the program's help text and trying each numeric option until one moves the column by exactly the predicted ratio; we tried this search only on HMMER and DIAMOND, and we have not evaluated its risks, such as an option like a random seed moving a column by coincidence. With the option set to the new release's value, the program prints on a subset the numbers it would print on the whole.

*No size dependence → no pin.* If every column fits identity, TARE replays without a pin. This is the case for Infernal under Rfam's recommended command, which derives its search size from the query sequence instead.

*Side files and competing entries.* Rfam's recommended command passes a "clan" file listing families that compete. On a subset, TARE rewrites that file to the families present, because the program rejects names it cannot find. If any member of a declared group changes, TARE re-searches the whole group, because a change to one member can reveal a hit of another that the cache never recorded.

The gate applies three rules in order and stops at the first that fires:

$$D = \begin{cases} \text{REFUSE} & \text{no contract} & \text{(R1)} \\ \text{REFUSE_PIN} & \text{size-dependent, no pin} & \text{(R2)} \\ \text{REFUSE_PIN} & \text{in-situ check fails} & \text{(R3)} \\ \text{AUTHORIZE} & \text{otherwise} & \end{cases} \quad (1)$$

R1 fires when the output does not split or a column fits no hypothesis. R2 fires when a column depends on the database size and no pin is available; TARE never guesses. R3 replays the probe sample against its own hits with the pin set and requires every byte to match the whole run. Every refusal runs the program in full; REFUSE_PIN differs from REFUSE only in what it reports: the output does decompose, so declaring a working pin could enable replay.

### 4.4 Stage 4: Discovery

Changed entries may produce hits the cache has never seen. TARE runs the program on all records against the changed entries $\Delta$ alone, with the pin set. A record's *candidates* are its cached hits on stable entries plus its new hits:

$$C_q = \{e \in S : (q,e) \in f(Q,R)\} \cup \{e : (q,e) \in f_\pi(Q,\Delta)\}. \quad (2)$$

Discovery compares every record with every changed entry, so its cost is proportional to the changed fraction of the reference. It is the one cost TARE cannot avoid (Section 4.7).

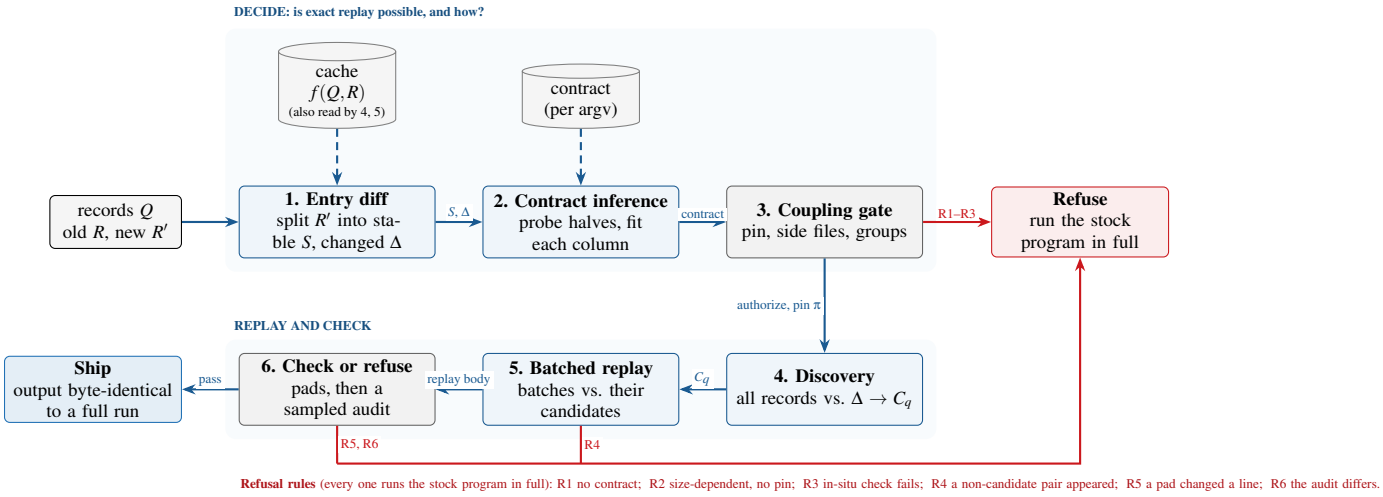



Figure 3: **How TARE updates an annotation.** Blue boxes do work; gray boxes are decision points. The top row decides whether the program's coupling can be neutralized; the bottom row reruns the unmodified program on candidates only and checks the result. Dashed arrows read stored state. Every refusal (red) leads to a full stock run, so a user always receives the program's own output.

## 4.5 Stage 5: Batched Exact Replay

Cached lines cannot be returned as they are, because their numbers were computed for the old release. TARE instead reruns the program on the candidates with the pin set, and the program prints the new numbers itself. Rerunning every protein against all candidates of all proteins would waste work: on one genome, the proteins' candidates together cover 3,719 families, but the mean protein has two. TARE therefore groups proteins into batches, runs each batch against only its own proteins' candidates, and stitches the outputs back together in the original order. It chooses the grouping that minimizes

$$T_{\text{replay}} = \alpha k + \beta \sum_b |U_b| + \gamma \sum_b |B_b|\,|U_b|, \tag{3}$$

where $k$ is the number of batches, $|B_b|$ and $|U_b|$ the proteins and families in batch $b$, $\alpha$ the cost of starting the program, $\beta$ the cost per family of preparing a sub-database, and $\gamma$ the cost of one protein–family comparison. Stitching is safe for HMMER because, in its source code, each protein's output block depends only on that protein's own hits, including the column widths and the per-protein statistics. As a tripwire, if a replay ever reports a protein–family pair outside the candidates, TARE refuses (R4), because a full run could contain pairs the cache never saw.

## 4.6 Stage 6: Check or Refuse

The replay assumes the program is entry-local. The check tries to prove that assumption wrong on the actual run. It reruns the replay with *pads*, families that were deliberately left out, added back to the subset. If the program is entry-local, pads change nothing: no candidate line changes and no pad family is reported. If anything changes, the replay depended on what it left out, and TARE refuses (R5).

TARE uses two kinds of pads. The *neighbor pad* adds, for every replayed protein, the families that sit next to its candidates in the database file; it catches neighbor coupling by construction, the failure we found in DIAMOND (Section 6.4). *Random pads* add three random samples of left-out families, each as large as the candidate set, for a fixed sample of $s = 50$ proteins. They test for dependencies of unknown form. If an unknown dependency affects $k$ of $n$ proteins and a random pad includes the responsible family with probability $q$, the check misses it with probability approximately

$$P_{\text{miss}} \approx h + (1-h)(1-q)^3, \qquad h = \binom{n-k}{s} \Big/ \binom{n}{s}, \tag{4}$$

where $h$ is the chance that the protein sample contains no affected protein. The approximation treats each pad as a single detection event; when several affected proteins fall in the sample, each pad has more chances to detect, so the true miss probability is lower. The neighbor pad needs no such formula: it contains every candidate's file-order neighbor by construction. The check is a test, not a proof. As a final safeguard, an audit reruns the stock program on a random sample of proteins, each included with probability 0.02 and at least 20 in total, and refuses if any line differs (R6).

## 4.7 Cost Model

Let $T_{\text{stock}}$ be the time of a full run on the new release and $f_\Delta = |\Delta|/|R'|$ the fraction of entries that changed. TARE's

update time is approximately

$$T_{TARE} \approx a f_\Delta T_{\text{stock}} + c\,T_{\text{stock}}, \tag{5}$$

where the first term is discovery ($a$ is near one if the program's time grows linearly with database size) and $c$ collects every other cost (replay, check, audit, parsing) as a fraction of a stock run. The speedup $1/(af_\Delta + c)$ can never exceed $1/f_\Delta$, and even for a tiny release it cannot exceed $1/c$. The first run also fills the cache, which costs a full run plus the contract probes; with fill time $F$ and update time $U$, TARE becomes cheaper than repeated full runs after $n \geq (F - T_{\text{stock}})/(T_{\text{stock}} - U)$ releases.

## 5 Implementation

TARE's core is 12,555 lines of Python, with 358 test functions. Its core contains no program names: small parsers read profile databases, covariance-model databases, FASTA files, and tabular outputs, and users declare pins, side files, groups, and reporting thresholds as data. The cache is a SQLite file keyed by entry hash and record. A typical invocation wraps the normal command shown below. The cache is namespaced by a hash of the command line with the reference path removed, so changing any option, including the thread count, starts a new cache and a new contract. This is conservative: repeated stock runs were byte-identical in every timed experiment. The namespace also includes a SHA-256 of the program binary, so upgrading a program in place starts a new cache.

```
tare run --strategy reference --pin '-Z {entry_count}' \
  --cache cache.sqlite -- hmmscan --cut_ga --noali --cpu 4 \
  --tblout {output:tblout} --domtblout {output:domtblout} \
  {reference} {input}
```

## 6 Evaluation

We ask four questions. **Q1:** Is TARE's output identical to a full run? **Q2:** How much faster is an update, and does the cost model predict it? **Q3:** Does TARE refuse when replay would be wrong? **Q4:** How does it compare with existing tools?

**Method.** Every experiment followed the same protocol, designed so that a failure means the method failed rather than that the test was blind. We first read the program's source for behaviors that could make a comparison fail for unrelated reasons, then ran a pilot on separate data, then committed the design and pass/fail rules to version control before the measured run. Each test included a case that must pass and a case that must fail, and a minimum number of output lines below which a result counts as inconclusive. Measured runs used genomes drawn by a fixed random seed and never used before. One rule changed after its pre-registration, during a measured run; we disclosed the change and reran every control (Section 6.4).

**Setup.** HMMER 3.4 (`hmmscan`) with Pfam 38.1 and 38.2; Infernal 1.1.5 (`cmscan`) with Rfam 15.0 and 15.1; DIAMOND 2.2.5 and BLAST+ 2.14.1 with Swiss-Prot 2026_01 and 2026_03; queries are bacterial genomes from NCBI RefSeq. Server timings use a dedicated AMD EPYC 7542 node; laptop timings use an Apple laptop shared with other work and are indicative only.

Table 2: Exactness experiments. "Lines": output lines compared byte for byte with a stock run on the new release (HMMER: hit plus domain table). "Differ": lines that differed. "Unpinned": the same replay without the pin. "Replay" runs the replay engine alone; "tool" runs TARE end to end, with batching and the check.

| Experiment | Lines | Differ | Unpinned differ |
|---|---|---|---|
| HMMER replay, 300 proteins | 1,079 | 0 | 534 of 534 |
| HMMER replay, 3 genomes | 51,406 | 0 | 26,094 of 26,127 |
| HMMER tool, 2 genomes (laptop) | 35,434 | 0 | not run |
| HMMER tool, 9 genomes (server) | 155,577 | 0 | not run |
| HMMER tool, 48 diverse genomes | 463,225 | 0 | not run |
| Infernal replay, 10 genomes | 4,217 | 0 | no pin needed |
| Infernal tool, 1 family removed | 459 | 0 | no pin needed |
| Infernal tool, 10 new genomes | 4,307 | 0 | no pin needed |
| Total | 715,704 | 0 | |

### 6.1 Metrics

**Exactness.** A run is EXACT if its body lines equal the stock run's as byte sequences; otherwise we count and classify the differing lines. Over $N$ compared lines with none differing, the one-sided 95% Clopper–Pearson upper bound on the per-line error rate is

$$p_{\text{up}} = 1 - 0.05^{1/N}. \tag{6}$$

Lines of one run share a pin and a reference, so they are not independent and the bound is descriptive, not a guarantee. **Speedup.** For genome $g$ with $r$ alternating runs per arm, $S_g$ is the median stock wall time divided by the median TARE wall time, both measured around the whole process. Across genomes we report the median of $S_g$ with a 95% percentile bootstrap interval, the geometric mean, and the range. Appendix A defines the remaining statistics (Wilson intervals, $t$-intervals for baseline overheads, coefficient of variation, model error, and safety outcomes).

### 6.2 Q1: Is the Output Exact?

Across the eight experiments in Table 2, not one of 715,704 lines differed from the stock run, so by Equation (6) a per-line error rate above $4.2 \times 10^{-6}$ would be unlikely to produce this result (one-sided 95%). Without the pin, nearly every line differed: 26,094 of 26,127 hit lines on three genomes ($d = 99.87\%$, Wilson 95% CI 99.82–99.91%) and all 534 on 300 proteins. The pin is therefore necessary and, with `--cut_ga`, sufficient.

**Diverse genomes and needless refusals.** On 53 further genomes (14 archaea, 30 bacteria, 2 eukaryotes, and 7 metagenome-assembled genomes), TARE shipped 48 updates, all byte-identical, and produced no unsafe result. It refused the other 5 although their replays matched stock: 3 because two domain-table rows of one protein shared the key (protein, family, E-value), and 2 because a text column held some values that parse as numbers. The *Drosophila* proteome, with 96 proteins of more than 20 domains, was one of the latter; three further eukaryotes did not finish within 12 hours and one bacterial job crashed on a storage error, so neither is counted. The duplicate-key rule also refused one of the ten server genomes below. These refusals cost speed, not correctness, and both are parser rules we can fix.

**Where exactness ends.** When HMMER reports hits by E-value threshold instead of `--cut_ga`, the replay matched on 300 proteins but failed on 1,000, differing on 3 hit lines and 5 domain lines. Hits near the threshold move in or out of a protein's list, which changes that protein's statistics; this mode is not entry-local and we do not claim it. The early pass on 300 proteins is why we require a minimum number of lines before accepting any result.

**A second program.** Infernal, which searches RNA families, behaved as its source code predicted: under Rfam's recommended command every numeric column fits identity, so no pin is needed. Replaying each genome's hits was byte-identical on ten genomes (4,217 lines) with and without clan competition. In that command, overlapping hits of competing families are marked rather than deleted, so removing a family cannot reveal a hidden hit; removing the family `GlmZ_SraJ` from one genome's reference was exact through the shipped tool (459 lines). With the non-default option `--oskip`, which deletes marked overlaps, the same removal was refused, because fourteen overlap marks depended on families that had no remaining line. Through the shipped tool on ten further genomes (Rfam 15.0→15.1, three alternating runs per arm), every one of the 60 updates was byte-identical to its paired stock run in both variants, with 4,307 body lines per variant against a pre-registered floor of 2,000.

## 6.3 Q2: How Much Faster?

Figure 4 shows that the replay and check designs decide whether TARE is faster than a full run. Our first version replayed every protein against all candidates and checked the result with four more full-size replays. On the laptop, with three alternating runs per arm, it took a median of 895 s against a median stock time of 680 s ($S_g = 0.76\times$; CV 3.7% for TARE and 1.7% for stock); the check alone took a median of 668 s. Batching made the replay 5.2× cheaper and sampling made the check 17.9–19.3× cheaper (ratios of the first version's median check time to each shipped run's). On two fresh genomes of 4,583 and 5,019 proteins, TARE replayed

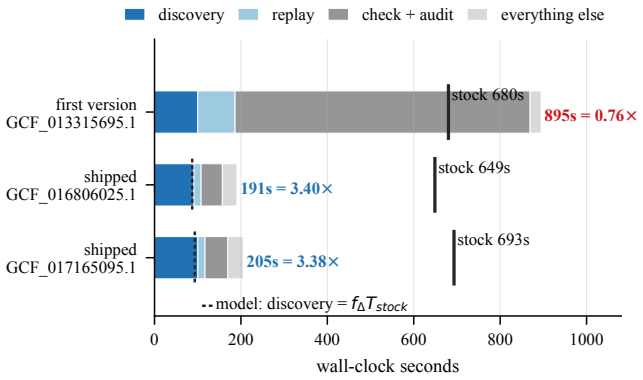



Figure 4: **Where the time goes** (laptop, indicative). Each bar is one update run, split into four stages; the black tick is a stock run on the same genome. For the first version (top), the run shown is the median of three alternating runs, with the matching stock run. The first version spent 668 s in its safety check and was slower than stock. Batching and sampling (bottom two, fresh genomes) cut the update to under a third of a stock run. Dotted ticks: the cost model's prediction for discovery.

in 107 and 111 batches and took 191 s and 205 s against stock runs of 649 s and 693 s: $S_g = 3.40\times$ and $3.38\times$ (geometric mean $3.39\times$), with byte-identical output. These are single runs on a shared machine, so we report no confidence interval for them; the server headline repeats each arm three times.

**The cost model.** With $f_\Delta = 0.134$, the discovery term predicts 87 s and 93 s on those genomes; discovery measured 91 s and 100 s, a relative error of 4.3% and 7.1%. The remaining costs, as a fraction of each stock run, average $c = 0.152$ (arithmetic mean of 0.153 and 0.151), so even a release that changed nothing could be at most $6.6\times$ faster with today's overheads (Figure 5); the check, audit, and parsing are therefore the next targets. The cache fill took 871 s and 852 s (single runs), 1.2–1.3× a stock run, so TARE pays for itself from the first update (projected from these indicative times).

**Other releases.** Strict hashes of all seven consecutive Pfam pairs (Figure 6) put the changed fraction between 2.0% and 93.1%. At those measured values the model predicts 3.5–$5.8\times$ for the four other pairs that changed at most 12.2%, $1.8\times$ for 37.0→37.1 (39.3%), and $0.88\times$ for 37.4→38.0, which rewrote 93.1% of Pfam's models. That release was a rebuild, not churn: 22,925 of its 22,965 edited models carry a new accession version, which `hmmscan` prints, and 19,976 have different model parameters. For a release like that, a full run is cheaper, and TARE does not yet decide this on its own. We have timed only the 38.1→38.2 pair; a cache carried across several releases is not yet measured.

**Server headline.** On a dedicated node, across nine fresh genomes with three alternating stock and TARE runs each, the median of $S_g$ is $4.24\times$ (95% percentile bootstrap CI 4.17–

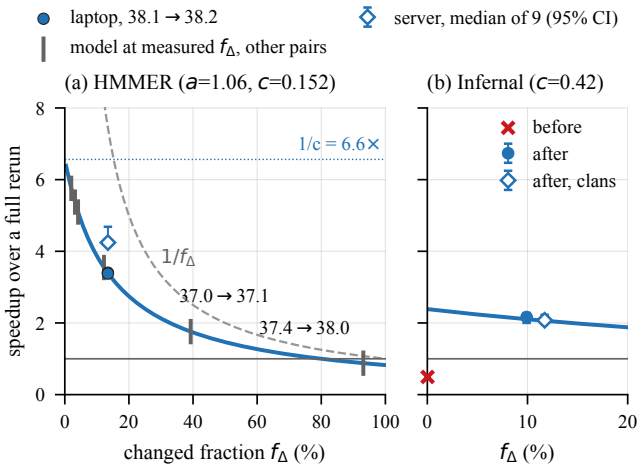



Figure 5: **The cost model across release sizes.** Speedup $1/(af_\Delta + c)$, with $a$ and $c$ measured on each program. (a) HMMER, with $a$ and $c$ from the two laptop updates: those updates (filled circles), the server median of nine genomes with its 95% CI (open diamond), the model at the measured $f_\Delta$ of the six other Pfam pairs (ticks), the bound $1/f_\Delta$ (dashed), and the overhead ceiling $1/c$ (dotted). (b) Infernal, with its own larger overhead: the first version (cross) and the shipped tool, medians of ten genomes with 95% bootstrap CIs.

4.68; geometric mean 4.39×; range 4.13–5.07×), and every update was byte-identical. The pre-registered draw had ten genomes: one refused at cache fill (the duplicate-key rule above) and is listed, not imputed, and three that hit the 4-hour job limit were rerun unchanged with 8 hours, a deviation we recorded before the rerun. The largest per-arm CV was 22.1% (flagged, not dropped, as pre-registered), and the pre-registered speedup prediction of 4.08× was within 19.4% of every genome's measured speedup. Filling the cache took a median 1.32× a stock run, so TARE pays for itself on the first update.

**Infernal, with a reduced check.** On Infernal, TARE ships with two of its four safety layers skipped on most inputs, and readers should weigh its Infernal speedups accordingly. The speedup did not at first carry over: only 457 of Rfam 15.1's 4,227 families (10.8%) changed, yet in the measured single-family removal the update took 666 s against a 328 s stock run, exact but 0.49×. A bacterial genome is one DNA sequence or a handful, so the audit's record sample and the random pads covered the entire genome, and an audit of a whole genome *is* a stock run. We therefore budget the audit and the random-pad sample by residues: when a single record holds more than half of the input's residues, those two layers are skipped and the output says so; the neighbor pad and the in-situ check always run. The change leaves HMMER's decisions and output byte-identical on a proteome, and every known-bad control still refused.

With this change, Infernal updates on ten fresh genomes ran

Table 3: Safety controls under the shipped check. Every case known to be wrong was refused, the correct case passed, and no run produced wrong output.

| Control | Expected | Result |
|---|---|---|
| DIAMOND, 6 lines lost | refuse | REFUSE |
| DIAMOND, 22 lines differ | refuse | REFUSE |
| 5% candidates dropped | refuse | REFUSE |
| HMMER genome (correct) | pass | pass, exact |
| DIAMOND, fresh sample | no wrong output | SAFE-REFUSE |
| BLAST+ boundary | refuse | REFUSE (R1) |

a median $S_g = 2.16\times$ faster than stock without clan competition (95% bootstrap CI 2.00–2.28; geometric mean 2.07×; range 1.41–2.37) and 2.07× with it (CI 1.97–2.23; geometric mean 2.10×; range 1.96–2.33), against cost-model predictions of 2.08× and 2.05×. These are laptop timings with three alternating runs per arm: the largest per-arm CV was 29% (flagged, not dropped, as pre-registered), and the largest per-genome prediction error was 32%.

**Infernal payback is slow.** Filling the cache took a median of 4.45× a stock run (range 3.85–5.35×), because the one-time contract probes and the in-situ check run on whole genomes; on the first genome, the probes took 609 s and the in-situ check 365 s, against a 325 s stock run. At the median speedup, the fill pays for itself only after about 6.4 updates. For Infernal, TARE is therefore worthwhile today only when the same genomes are re-annotated across several Rfam releases; cutting the fill is the main remaining work.

## 6.4 Q3: Does TARE Refuse When It Should?

**A failure, and what it revealed.** The check's design came from a failure. When we declared DIAMOND's database-size option (`--dbsize`) as a pin, the replay on 300 proteins lost 6 of 31,377 stock lines while every other line was byte-identical, and our first check, which used only random pads, passed it. Under our pre-registered rule, that was an UNSAFE result and invalidated that check. To find the cause, we added left-out entries back in halves until each missing line returned. Every one of the six lines was restored by exactly one entry: the candidate's neighbor in the database file. Figure 7 shows the mechanism in DIAMOND's source: its first filter reads bytes across the boundary into the preceding entry. DIAMOND's results therefore depend on the order of the database. A random pad contains one particular neighbor with probability 0.049, so three random pads miss it 86% of the time, which is why the first check passed.

Rebuilding the full Swiss-Prot database in five other orders, and searching the same 2,000 queries, changed that table. Across the five orders, 173 lines disappeared, 148 appeared, and 1 line reported a different alignment, out of 210,460 stock lines. Of the 322 changed rows, 321 had an alignment inside

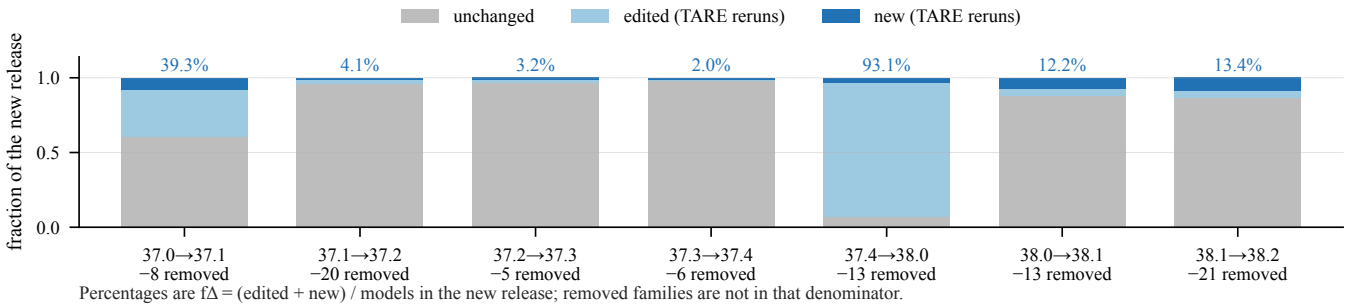



Figure 6: **Edits, not just new families, decide how much an update reruns.** Strict hashes of seven consecutive Pfam releases. Gray is unchanged. Light blue is edited and dark blue is new; both are what TARE reruns. The label is $f_\Delta = (\text{edited} + \text{new})/Z_{\text{new}}$. Pfam 37.4→38.0 reruns 93.1% of the new release: 22,965 edited and 822 new, with 1,758 unchanged, of 25,545. Pfam 38.1→38.2 reruns 13.4%: 1,378 edited and 2,674 new, with 26,082 unchanged, of 30,134, the same pair as Figure 1. The smallest change is 37.3→37.4, at 2.0%.

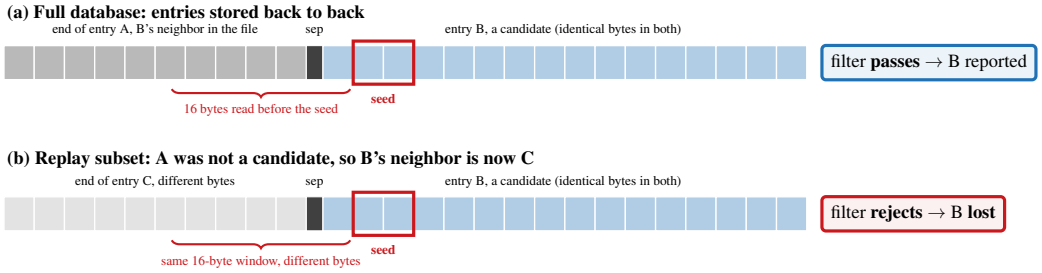



Figure 7: **Why DIAMOND cannot be replayed exactly.** DIAMOND stores database entries back to back, separated by a single byte. Its first filter examines a window of 16 bytes before and 32 after each short matching "seed" without stopping at the entry boundary (DIAMOND 2.2.5, `finger_print.h`). A seed near the start of entry B therefore reads the end of whichever entry precedes B. Removing that neighbor from the database changes the window and can drop B's hit. The neighbor pad adds A back, the file-order neighbor of every candidate; B's line then reappears, the check sees a line change, and TARE refuses (R5). Each of the six lines DIAMOND lost was restored by exactly one such neighbor.

the filter's window of the entry boundary. We have not shown that a command-line option removes this dependence.

**The shipped check.** Adding the neighbor pad closed this gap (Table 3). On a fresh sample of 2,000 proteins, the replay differed from stock in 22 of 227,223 lines (20 missing, 2 extra), (0.0097% of lines), and the check failed on all five pads: a true SAFE-REFUSE. A second fresh DIAMOND sample was also refused. BLAST+ was refused earlier, at contract inference: one of 15,125 aligned lines has a percent identity (column 2) that matches no normalizer, so R1 fired. That line is a different alignment, measured below. In Table 3 the correct HMMER case passed; across all HMMER updates, 6 of 63 were refused needlessly (Section 6.2).

The mismatched BLAST line is a different alignment: the whole-database row prints identity 33.333 and the half-database row prints 21.875 for the same query and subject. Pinning `blastp` with `-dbsize` set to the letters in the new database makes both halves byte-identical to the rows the whole database produced for those subjects, 7,498 lines and 7,627 lines, together the 15,125-line table. A pilot of that pin was byte-identical to stock on 15,498 lines, and dropping 748 subjects removed 780 stock rows. The shipped command still refuses, because duplicate rows have no row-index column, so BLAST is not a program TARE ships.

**Limits.** For dependencies other than file-order neighbors, the random pads are weak: with 22 affected proteins of 2,000, the shipped sample of 50 misses with probability 0.71 by Equation (4). Every control in Table 3 was refused or passed as expected, but that does not make the check complete, as the next paragraph measures.

We measured that weakness by injecting coupling into a stand-in for HMMER, a synthetic scorer that reads the same profile database and FASTA and prints `hmmscan`-format tables, so each trial takes milliseconds: 1,400 trials on 2,000 records and 2,500 models (Figure 8). With no injected coupling, all 200 trials shipped. When the printed note depended on the whole searched file, a length sum or a hash of every model, the neighbor pad refused all 200 trials of each form. When a hit depended on one hidden model that is not a file-order neighbor, or on a model 2, 5, or 20 positions away,

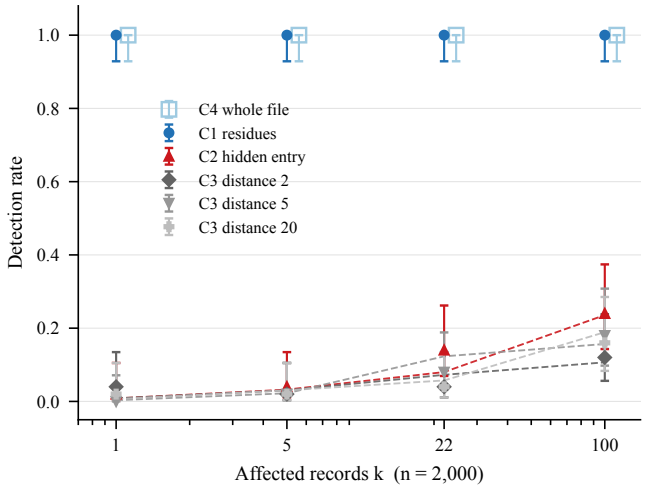



Figure 8: **The random pads miss coupling that is not a file-order neighbor.** Each point is the fraction of trials the check refused, with a Wilson 95% interval. Dashed curves are the random-pad formula (Equation (4)) for the forms it describes. File-wide coupling sits at the top of the plot; hidden and distant coupling does not. With no coupling (C0), 0 of 200 trials were refused. C4's squares are offset slightly right so both markers at 1.0 stay visible. The audit did not run in these trials.

the check missed the fault on 741 of 800 trials and refused 59. In the distance-2 cell with $k = 1$, the miss formula's detection is 0.008, below the Wilson 95% interval's lower bound of 0.011. One real `hmmscan` run of each form, not part of these rates, shipped the clean case and refused the four coupled cases.

Each grid trial started from an empty cache, so no cached rows were reused and the audit (R6) never ran; these rates therefore measure the pads alone. In a separate run on a warm cache, the audit refused a fault the pads had missed. Equation (4) held up: in 8 of the 16 cells it describes, measured detection was below its prediction, each within the 95% Wilson interval, and the one cell outside its interval detected more than predicted. For a program TARE has never seen, these numbers mean hidden coupling that reaches only a few records is likely to ship; the guarantee holds when the program is entry-local or its coupling is file-wide or between neighbors.

**A disclosed change.** During the measured run of the sampled check, we changed one rule so that a candidate of one protein that is also a neighbor of another counts as a candidate rather than a pad hit, after it caused a needless refusal. Because that change made the check more permissive, we disclosed it and reran every control under the final code. Every known-bad case was still refused, for the same reason as before.

**What the option search cannot do.** On HMMER, TARE's option search found `-Z` from probe runs alone, but it cannot infer reporting thresholds, so users declare them. On DIAMOND, it found no size option. Neither led to wrong output, because a missing pin is a refusal (R2).

Table 4: General-purpose caches on HMMER: what each does on new input, and its first-run overhead against stock (mean, 95% $t$-interval).

| System | New input | First-run overhead |
|---|---|---|
| Riker [7] | reruns | 1.62× [1.53, 1.71] |
| INCR [28] | reruns | 2.24× [2.06, 2.43] |
| ProcessCache [24] | reruns | not built (x86 only) |
| TARE | replays changed part | 1.32× (server median) |

## 6.5 Q4: Comparison with Existing Tools

**General-purpose caches.** Riker, INCR, and ProcessCache all re-executed HMMER on a new genome, as their designs require, because the input bytes changed (Table 4). Riker skipped an identical rerun in the same container; INCR re-executed even an identical rerun, because it records the command's output file as a dependency that does not yet exist and rejects it once the file exists. Tracing also costs time on the first run. We measured it as the arithmetic mean of $n = 4$ paired time ratios against stock in one uniform container (500 proteins, all 30,134 families): Riker took 1.62× the stock time (95% CI 1.53–1.71) and INCR 2.24× (95% CI 2.06–2.43); both intervals lie above the stock time. TARE's first-run cost in Table 4 is its HMMER cache fill on the server (median of nine genomes; on the laptop, 1.2–1.3×). ProcessCache builds only for x86 processors, so it has no overhead number in that ARM container; its rerun behavior was observed on the x86 server. These systems are correct for what they are designed to do; a database release is simply outside it.

**Approximate incremental search.** Specialized incremental search does reuse work across a release, but not exactly. We ran iSeqSearch's incremental DIAMOND [29] on the same 300 proteins as our DIAMOND experiment, updating Swiss-Prot from 2026_01 to 2026_03, and compared it with a full run (Figure 9). Its E-values are close (over 31,377 shared lines, Pearson $r = 0.999999$ and Kendall $\tau_b = 0.99999$), but only 407 of those lines (1.30%, Wilson 95% CI 1.18–1.43%) are byte-identical; 30,968 report the same alignment with different printed numbers, 2 report a different alignment, and 133 lines are hits to sequences the new release removed. iSeqSearch is designed for databases that only grow, and Swiss-Prot also removes entries, so this update lies partly outside its design; the stale hits are nonetheless wrong output for a user of the new release. TARE refuses this program and returns the full run, so it gives no speedup here; where it replays, every line is identical.

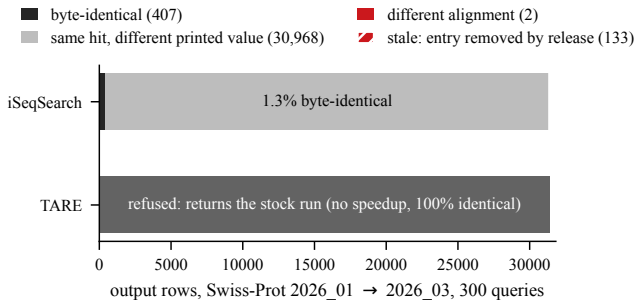



Figure 9: **Approximate versus exact.** Output of iSeqSearch's incremental DIAMOND after a Swiss-Prot update, compared line by line with a full run. Its numbers are close, but only 1.3% of lines match exactly and 133 lines report sequences the new release deleted. TARE refuses this program and returns the stock run, so it is exact but no faster; we did not time iSeqSearch.

## 7 Discussion

**Threats to validity.** *Generality:* the two programs TARE replays exactly, HMMER and Infernal, come from one research group and share code, so they may share the property that makes replay work; the two refused programs come from other groups. We therefore claim that the method works for programs whose coupling lies in settings they expose, and that it detects when coupling lies elsewhere; we do not claim it works for most search programs. BLAST's size pin reproduces a whole-database table and the shipped gate still refuses it, so BLAST is not a second shipped program, and an exactly replayed program from an unrelated codebase remains open. *Release choice:* Figure 6 measures seven consecutive Pfam pairs. The fraction that must be rerun ranges from 2.0% (37.3→37.4) to 93.1% (37.4→38.0). We have timed only 38.1→38.2, where that fraction is 13.4%. *Timing:* laptop numbers are indicative runs on a shared machine; the headline is measured on a dedicated node with alternating runs. *Check strength:* random pads miss hidden coupling, and on that grid most trials shipped a wrong replay (Section 6.4).

**Anticipated objections.** *"An expert could write this in a few lines of shell: run the changed families with `-Z`, then rerun each protein's candidates."* For HMMER with `--cut_ga`, yes, and that script is TARE without its gate, check, and audit. Removing those stages from the measured laptop runs bounds what it would gain: the check and audit took 7.6% of a stock run, so the script would reach about 4.6× where TARE reached 3.4× (an estimate from the measured stages, not a separate run). What the script cannot do is notice when its assumptions fail: run on DIAMOND, the same script silently drops hits, which is the case TARE refuses. *"Why not just rescale the cached E-values?"* Rescaling reproduces the program's arithmetic but not its rounding, and it must be written for each program and column; pinning lets the program print its own numbers.

Rescaling the cached E-values on the five HMMER proteomes in the exactness experiments matched 25,597 of 44,132 hit lines (58.0%, Wilson 95% CI 57.5–58.5%). No line was missing or extra. The mismatches are value-only: 10,926 full-sequence E-values and 10,878 best-domain E-values. At full-sequence E-value cutoffs of $10^{-3}$, $10^{-5}$, and $10^{-10}$, 0, 2, and 5 hit lines would cross the cutoff. We then ran the expert script itself. On the two HMMER genomes in Table 2 its tables were byte-identical: 8,714 and 9,291 hit lines, and 8,405 and 9,024 domain lines. On the 2,000-protein DIAMOND sample it silently dropped 68 of 227,223 stock lines, added 2, and changed 1 alignment. The speedup estimated by dropping the check and the audit remains an estimate from those stages; these runs were not timed on the server. *"InterProScan already reuses precomputed results."* Its lookup is keyed by the protein sequence for a fixed database version; it answers "has this protein been seen?" rather than "what changed in the database?" *"Use a faster or GPU search program."* That is complementary: it shrinks $T_{\text{stock}}$, and TARE's savings apply on top whenever the program's coupling can be pinned.

**Deployment.** The natural deployment is InterProScan [12] after a member-database update. Member databases that score each model against its own threshold, as Pfam does, should behave like our HMMER results; those that compare hits across models would need declared groups or would be refused. Pfam's own pipeline, `pfam_scan.pl`, adds clan competition after `hmmscan`; we evaluated clan competition only for Infernal, so that step is not covered by our HMMER results.

## 8 Related Work

**General-purpose incremental execution.** Build systems and workflow managers avoid repeated work by keying a command on the bytes it reads (Table 5). Riker traces system calls to rerun only commands whose inputs changed [7]; ProcessCache caches process executions [24]; INCR memoizes shell command effects and can split inputs into annotated chunks [28]; Nextflow, Snakemake, and Cromwell hash task inputs [3, 10, 17]; and Mokhov et al. model the design space [16]. Their unit of reuse is the command, so they cannot reuse part of an output whose numbers depend on the whole database. TARE is complementary: it decides what to rerun inside one command.

**Incremental sequence search.** iBLAST searches only appended sequences and corrects BLAST's E-values with a formula derived for BLAST [9]. iSeqSearch applies that correction to other programs with BLAST-style output, including DIAMOND [29]. GeStore runs unmodified BLAST and HMMER through administrator-written plugins that rewrite inputs and merge outputs [20]. MMseqs2 supports incremental clustering [25]. mpiBLAST splits a database across machines and fixes the global database size so that every piece prints the E-values of a whole-database search [8]; TARE borrows that

Table 5: Prior work on the problem of updating annotations after a database release. "Unmodified": runs the program as is. "Release-aware": reuses work when the database changes. "Identical": output equals a full run byte for byte. "Per-tool code": requires code or formulas written for each program. †TARE's pins, side files, groups, and reporting thresholds are declared as data on the command line; its core has no program-specific code.

| System | Grain | Unmodified | Release-aware | Identical | Per-tool code | Limitation for this problem |
|---|---|---|---|---|---|---|
| Riker, ProcessCache, INCR [7,24,28] | command / chunk | yes | no | yes | no | a release reruns everything |
| Nextflow, Snakemake [10,17] | task | yes | no | yes | no | task inputs change |
| iBLAST [9] | appended sequences | yes | growth only | rescaled | yes | BLAST-specific E-value formula |
| iSeqSearch [29] | appended sequences | yes | growth only | rescaled | yes | 1.3% of lines identical; stale hits |
| GeStore [20] | plugin | yes | yes | rescaled | yes | administrator-written plugins |
| Precomputed lookups [4,12] | query sequence | yes | no | yes | no | keyed by query, fixed release |
| KumQuat [23] | input stream | yes | no | yes | no | no database argument |
| IVM, self-adjusting [1,2] | operator | no | yes | yes | n/a | computation must be rewritten |
| mpiBLAST [8] | database segment | yes | no | yes | yes | parallel search of one release |
| **TARE** | **record × entry** | **yes** | **yes** | **yes** | **no**† | **refuses programs with hidden coupling** |

pinning and contributes what surrounds it: deciding which entries to rerun after a release, rerunning each record only on its candidates, and refusing when a check finds coupling the pin does not cover. InterProScan and eggNOG-mapper reuse precomputed matches keyed by protein sequence [4,12]. These approaches are hand-built for one program, approximate when the database changes, or keyed by query rather than database. Our DIAMOND result applies directly to them: no correction of DIAMOND's printed numbers can recover hits its filter dropped because a neighboring entry was absent.

**Black-box decomposition and incremental computation.** KumQuat synthesizes combiners that let an unmodified command run on pieces of its input stream [23]; PaSh classifies shell commands by how they parallelize and supplies aggregators for their outputs [27]; and Caruca classifies opaque commands by whether they keep state [14]. Incremental view maintenance and self-adjusting computation keep results current under input changes but require the computation to be written in their framework [1,2]. These methods split an input stream or a known program; TARE splits a database argument whose size enters every printed number, for a binary it cannot inspect. Its partition probes borrow the spirit of metamorphic testing [6,22] rather than claiming a new testing method.

## 9 Conclusion

Reference databases change every few weeks to months, and the programs that search them start from scratch every time. We showed that for programs whose cross-entry coupling lies in settings they already expose, this is unnecessary: pinning those settings lets an unmodified program reprint, byte for byte, the output of a full run while searching only what changed, and a check that adds back left-out entries catches coupling the pin does not cover, as it did for DIAMOND, whose filter reads neighboring entries. The same check misses most couplings that are not file-order neighbors. Every replayed line across HMMER and Infernal in Table 2 was identical to a full rerun. Across seven Pfam releases the fraction that must be rerun ranges from 2.0% to 93.1%; we have timed the 13.4% pair, and that update ran 4.24× faster than a full rerun on a dedicated server (median of nine genomes). An Infernal update ran about twice as fast with a reduced check. The next steps are timing the other releases in Figure 6, an unrelated program that ships, a check that bounds hidden coupling, a cheaper cache fill for Infernal, and a deployment inside InterProScan.

## Availability

An anonymized artifact accompanies this submission. It contains TARE, every pre-registration and addendum, the result files from which each number in this paper is taken, and the scripts that regenerate every figure from those files. The repository URL is omitted for double-blind review.

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# A Statistical Definitions

"Mean" is the arithmetic mean $\bar{x} = \frac{1}{n}\sum_i x_i$; "median" is the sample median; the sample standard deviation $s$ uses $n-1$ in its denominator; and "geometric mean" is $\left(\prod_{i=1}^{G} x_i\right)^{1/G}$, used only to summarize ratios. Differing lines are classified as whitespace-only, value-only, missing, extra, or reordered.

**Wall time.** $T$ is the elapsed wall-clock time of one complete invocation, measured with a monotonic clock around the process, including start-up, parsing, and all program calls. Database preparation (`hmmpress`) before either arm is excluded from both.

**Speedup.** $S_g = \text{median}_{i\le r}\, T_{g,i}^{\text{stock}} \big/ \text{median}_{i\le r}\, T_{g,i}^{TARE}$, a ratio of medians, which is robust to a single slow run. On the laptop $r = 1$ for HMMER and $r = 3$ for Infernal; on the server $r = 3$. The pre-registered server draw is $G = 10$ genomes and 10,000 bootstrap resamples. Run-to-run variability per arm is the coefficient of variation $\text{CV} = s/\bar{T}$.

**Pin necessity.** For the unpinned control, the fraction of differing lines $d = D/N$ is reported with a 95% Wilson score interval.

**Overhead of other tools.** Each paired run $j$ gives $\rho_j = T_j^{\text{tool}}/T_j^{\text{stock}}$; we report $\bar{\rho}$ and the 95% $t$-interval $\bar{\rho} \pm t_{0.975,n-1}\, s/\sqrt{n}$.

**Prediction error.** For a predicted value $\hat{x}$ and a measured value $x$, $\varepsilon = |\hat{x} - x|/x$; the text names the quantity (a stage time for discovery, a speedup for the headline and for Infernal, as pre-registered); we report the maximum over genomes.

**Agreement of approximate output.** For a tool that is not exact, we report the fraction of shared lines that are byte-identical (with a Wilson interval) and the agreement of its E-values with stock over shared lines, using Pearson's $r$ and Kendall's tie-corrected rank correlation $\tau_b$.

**Safety outcomes.** Each safety run is EXACT (replay equals stock and every check passed), SAFE-REFUSE (a check failed and TARE ran the program in full; *true* if the replay differed from stock, *needless* if it did not), or UNSAFE (replay differed and every check passed). Any UNSAFE result was pre-registered as a failure of the safety claim.